

AUBREY HAYES

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Education

University of California - Santa Cruz

June 2024 – June 2028

Bachelor of Science in Computer Science and Applied Mathematics

GPA: 4.0

Relevant Coursework

- Data Structures
- Algorithms
- Programming Abstr.
- Prob. & Statistics
- Computer Systems
- Machine Learning
- Discrete Math
- C Programming

Experience

Tech4Good Lab

March 2025 – Current

Director of Web Development

Santa Cruz, CA

- Developed and deployed a custom web-hosted IDE directly into the platform, supporting 30+ concurrent users through interactive web development guides and real-time practice activities.
- Engineered core features for the Causeway educational platform using Angular and Firebase, implementing NgRx and RxJS to handle application state and user interactions.
- Directed an agile engineering teams of 8+ full-stack developers, managing technical roadmaps, leading sprint planning, and conducting rigorous code reviews.
- Contributed to Human-Computer Interaction (HCI) research by facilitating qualitative user studies, performing literature reviews, and executing technical writing to drive user-centric platform improvements.

UCSC Humanities Division

September 2025 – Current

Web Assistant

Santa Cruz, CA

- Engineered the infrastructure recovery of a critical research platform, resolving persistent downtime by implementing an Apache reverse proxy and containerizing databases via Podman.
- Managed campus server configurations to resolve routing redirects, and refactored legacy codebases to securely integrate modernized Authentik SSO authentication flows.
- Led development across 15+ department and faculty sites, engineering reusable UI components and enforcing strict web accessibility compliance.

Google Developer Group

October 2024 – May 2025

Full Stack Developer

Santa Cruz, CA

- Architected a full-stack student resource prototype using React.js, Node.js, and Firebase during a 7-month collaborative development cycle.
- Designed scalable Firestore database schemas and integrated Firebase Authentication to handle real-time event data synchronization.

Projects

Educational MCP Server | *Google Cloud, Google Run, Firebase, TypeScript*

July 2026

- Engineered an MCP integration for AI assistants to autonomously query educational concepts and step-by-step guides, reducing friction in the learning process.
- Designed and integrated a backend logging service with Firebase to capture student errors and struggles, creating a feedback loop to improve curriculum design.

Multilingual Hispanic Speech Corpus | *Astro, NocoDB, Authentik, Bun*

December 2025

- Deployed a server-rendered Astro frontend and Bun backend supporting data access for cross-institutional researchers.
- Managed virtual machine environments and integrated Authentik to securely handle unique user accounts across UCSC, UC Berkeley, and UCLA.

Technical Skills

Languages: Python, C, HTML/CSS, JavaScript, TypeScript, SQL

Developer Tools: VS Code, Git, GitHub, Firebase, NPM, Docker, Podman

Technologies/Frameworks: Linux, Node.js, Angular, React, Astro, NgRx, RxJS, WordPress